

Logistic Regression (scored tasks: 4 points)

Tasks (Lab 4):

1. Problem of linearly separable classes.

Dataset *earthquake.txt* corresponds to the problem of predicting seismic shocks (volcanic eruptions and nuclear explosions) (variable **popn**) based on two variables: **body** (deep wave magnitude) and **surface** (surface wave magnitude).

- Make a scatterplot for variables **body** and **surface**. Mark classes corresponding to observations.
- Fit logistic model without regularization, print estimated coefficients, estimated probabilities and compute log-likelihood function.
- Fit logistic model with ℓ_2 regularization, print estimated coefficients, estimated probabilities and compute log-likelihood function.

Compare the two fits and explain what happens when the classes are linearly separable. Why can the coefficients in the unregularized model become very large?

To the scatterplot, add the decision boundary for both models.

2. Simulation example.

- Generate data from logistic model:

$$y_i \sim \text{Bern}(p_i),$$

where

$$p_i = \frac{1}{1 + \exp[-(\beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \dots + \beta_5 x_{i,5})]},$$

for $i = 1, \dots, n$, $x_{i,1}, \dots, x_{i,5} \sim N(0, 1)$, $n = 50$. Parameters: $\beta_0 = 0.5$, $\beta_1 = \dots = \beta_5 = 1$. Fit a logistic regression model (with intercept) and calculate the estimators of the coefficients $\hat{\beta} = (\hat{\beta}_1, \dots, \hat{\beta}_5)$. Repeat the experiment $L = 100$ times and estimate the MSE (mean squared error):

$$MSE = E(\|\hat{\beta} - \beta\|^2),$$

$$\widehat{MSE} = \frac{1}{L} \sum_{i=1}^L \|\hat{\beta}^{(i)} - \beta\|^2,$$

where $\|\cdot\|$ is Euclidean norm, $\beta = (\beta_1, \dots, \beta_5)$ is vector of true parameters, $\hat{\beta}^{(i)}$ is a vector of estimated parameters in i -th experiment.

- Repeat the experiment for $n = 50, 60, 70, 80, 90, 100, 200, 300, \dots, 1000$ and make a plot showing how $\widehat{MSE}$ depends on n .
- Using the same simulated datasets (the real model has $\beta_0, \beta_1, \dots, \beta_5$), train the model based only on 3 variables (and intercept): $x_{i,1}, x_{i,2}, x_{i,3}$ and draw an analogous curve showing how the $\widehat{MSE}$ for $\beta = (\beta_1, \beta_2, \beta_3)$ depends on n . Compare both curves and comment on the effect of fitting a misspecified model with omitted relevant variables.